

The Mystery of Fire

What does a flame need to keep burning?

The Role of Air in Burning Things



What does a flame need to keep burning?

Air is all around us — invisible, but powerful.



Air is everywhere.

You cannot see it. But can something invisible affect a flame?



Before



Revealed

FOCUS QUESTION

Can air affect fire?

Let's test it with a careful classroom experiment.



Build the investigation.

Board · Clay · Candle · Bottomless glass jar · Metal lid · Lighter (teacher only)

Board

Clay

Candle

Jar

Lid

Lighter+



Caution: Only the teacher should use the lighter to light the candle.

What do you think will happen?

If we seal the jar with a metal lid...

A

The flame keeps burning the same.

B

The flame becomes weaker, then may go out.

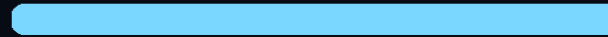
C

The flame goes out right away.

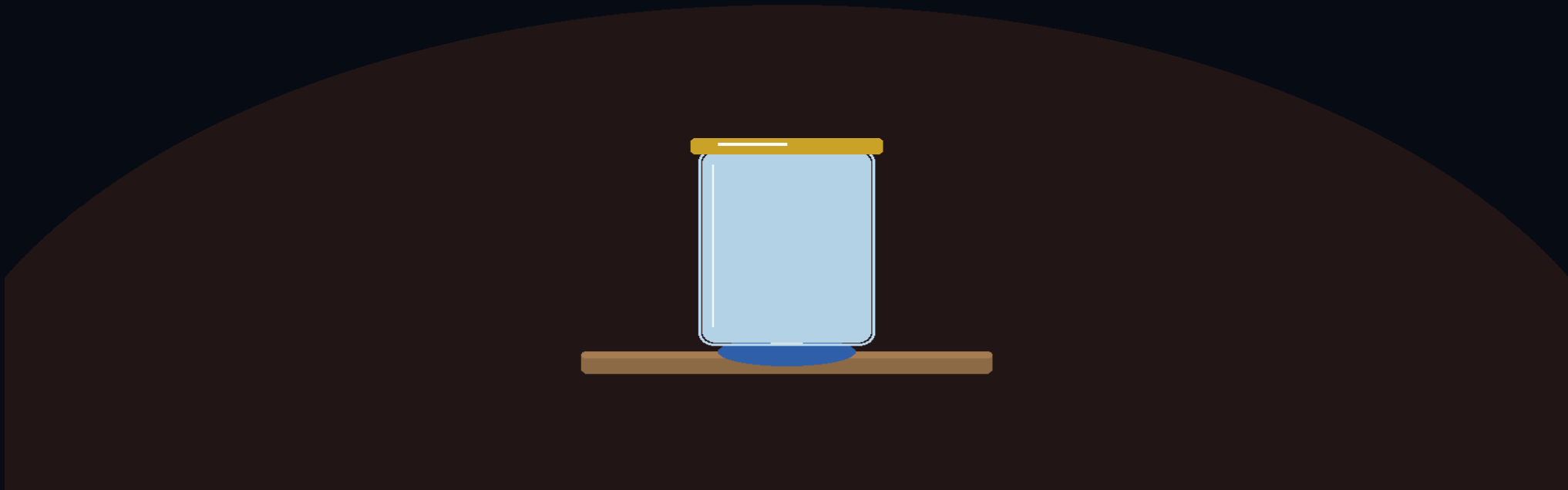
Vote as a class — then continue to test.

Trap the air.

Place the metal lid. Watch the air inside — and the flame.

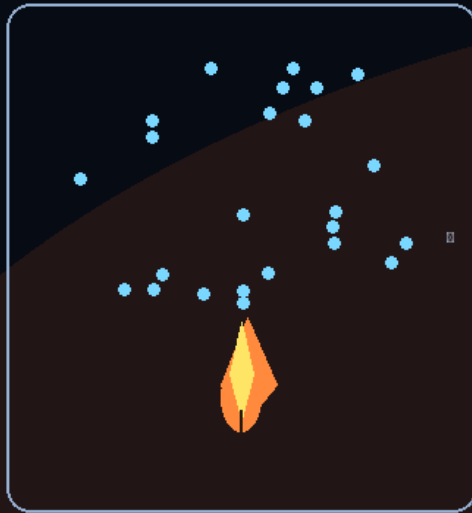


Available air inside the jar

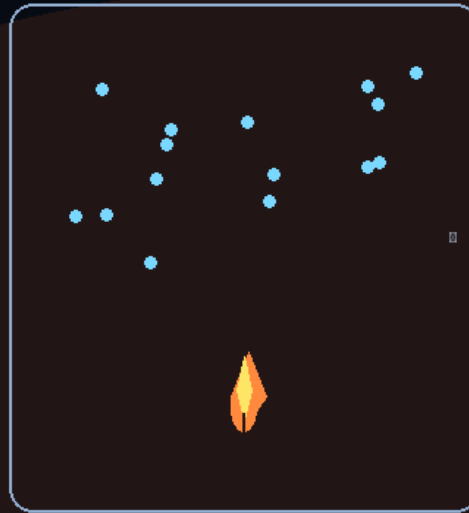


The flame goes out.

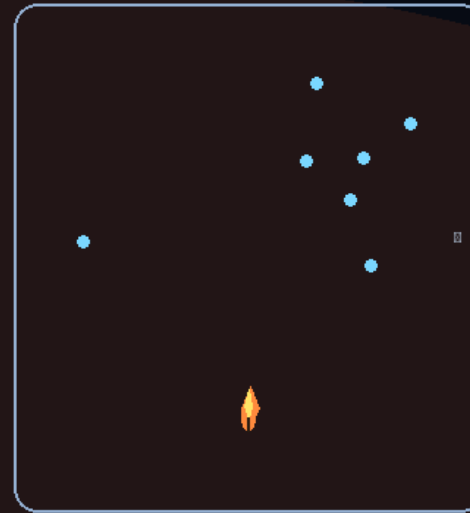
The flame gradually weakens... then the candle goes out after a short time.



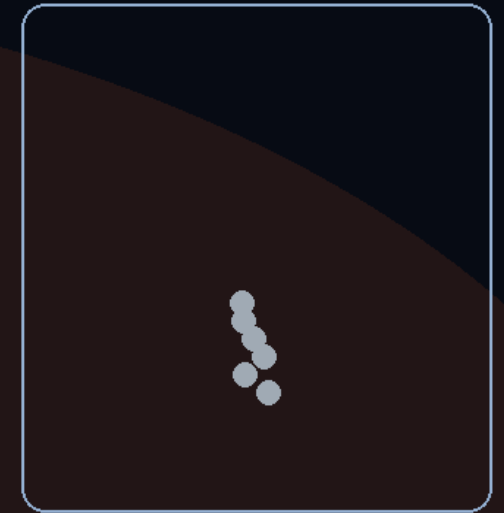
Air full



Air used



Almost gone

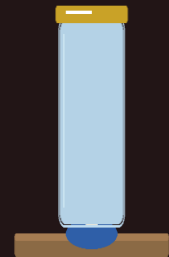
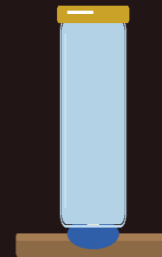
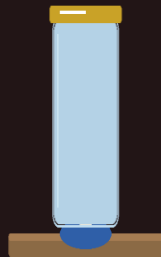
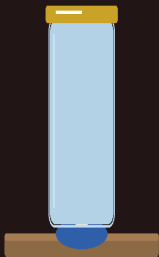


Flame out

Prediction check: B matches the observation best.

The air inside was used up.

In a closed jar, the air is used up. Once it runs out, the candle goes out.

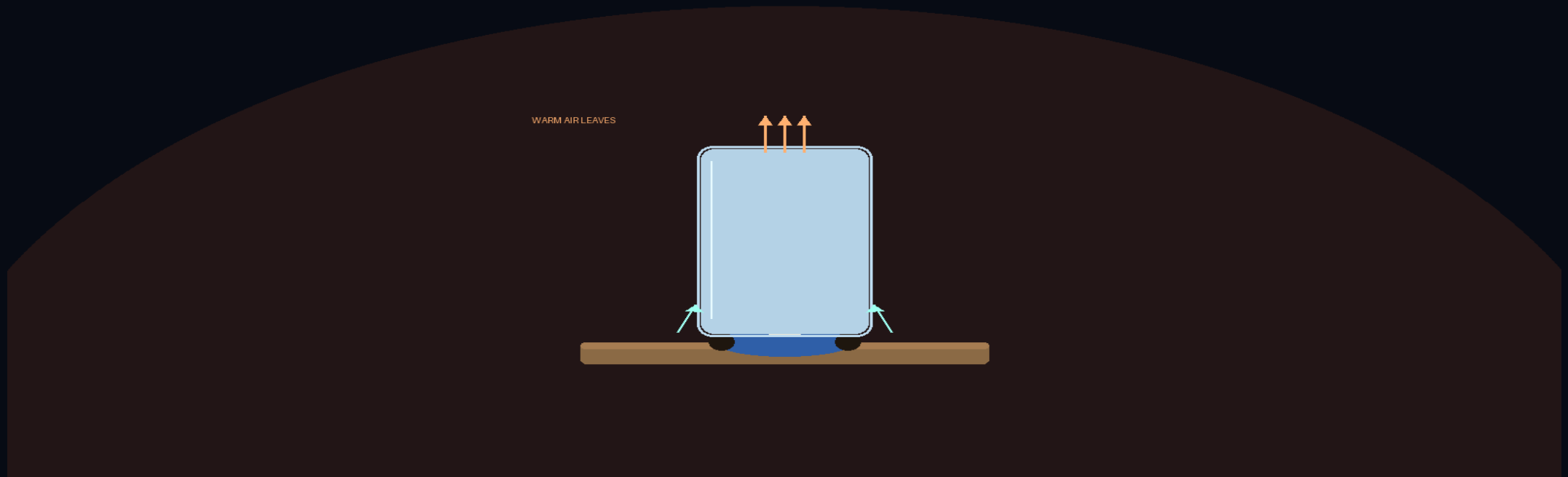


Fresh air in. Warm air out.

Gap at the bottom + opening at the top → air is constantly renewed.

FRESH AIR ENTERS

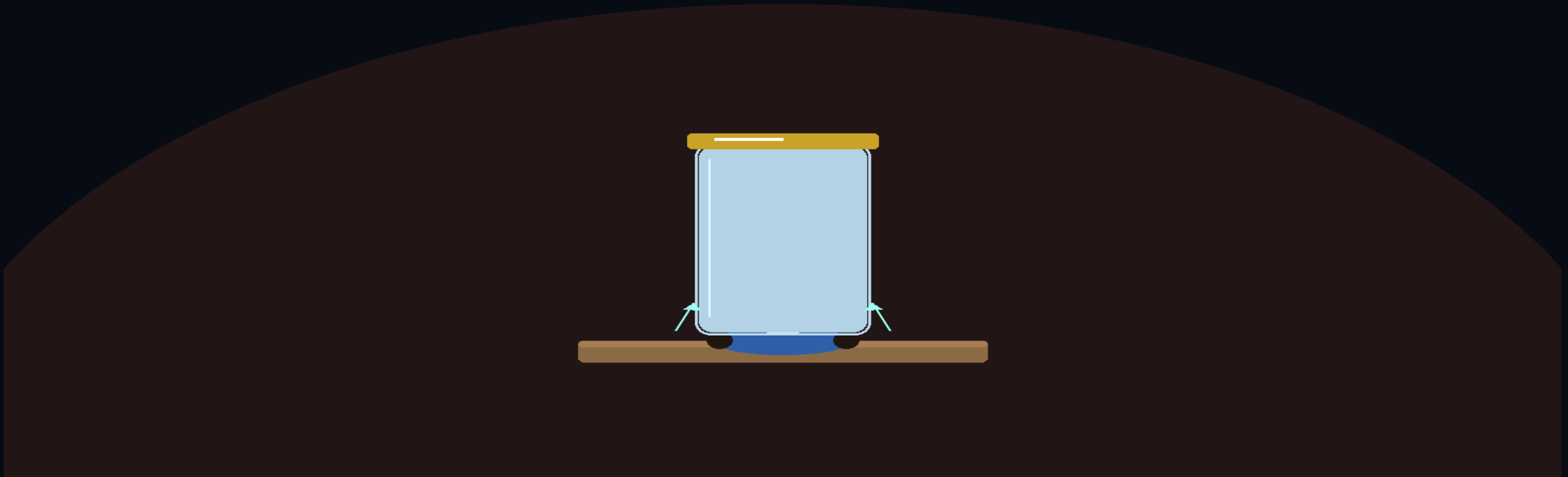
WARM AIR LEAVES



Bottom gap alone is not enough.

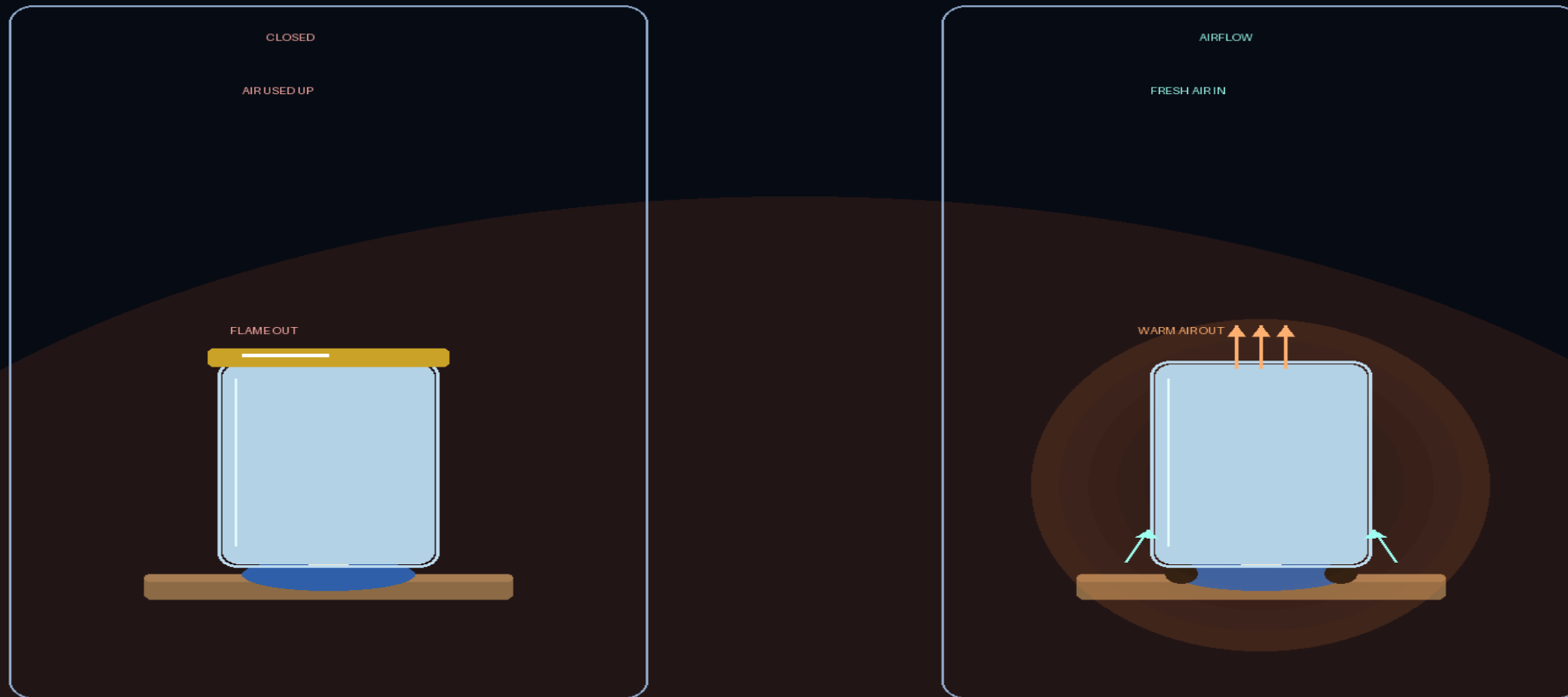
Lid on + small bottom gap → flame goes out after a slightly longer time.

Bottom gap only & flame still goes out



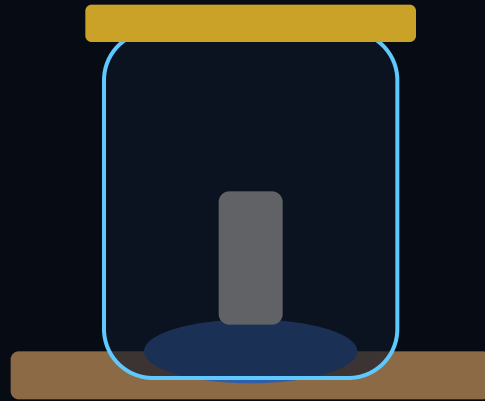
Closed vs. airflow.

Sealed air runs out. Continuous fresh air keeps burning going.

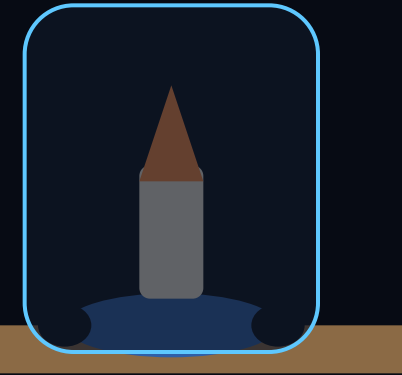


Build the science with shapes.

Closed (left) vs airflow (right) — PowerPoint objects you can edit.



CLOSED



AIRFLOW

What is meant by burning?

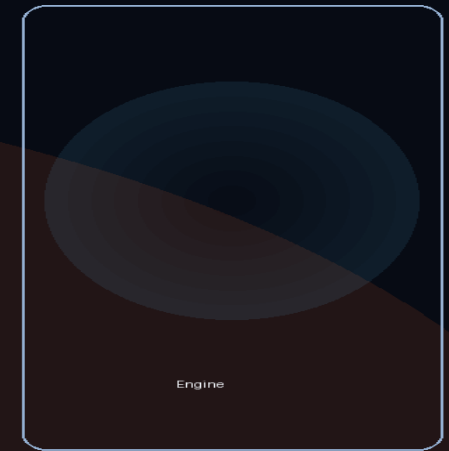
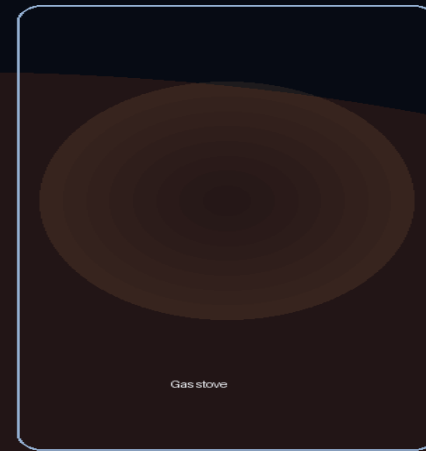
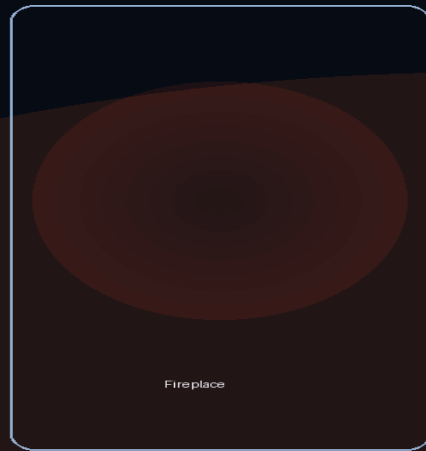
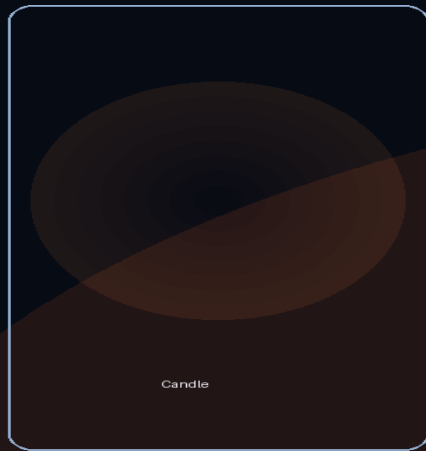
A reaction between oxygen and a substance that produces heat and light.



Burning (combustion): oxygen + a substance → heat and light

Where do we see this?

Burning needs a continuous supply of fresh air.



Candle · Fireplace · Gas stove · Engine

Which candle burns longer?

Choose carefully — then reveal.

A

Candle A — jar sealed with a lid, no gaps.

B

Candle B — lid on, but a small gap only at the bottom.

C

Candle C — gap at the bottom and open at the top.

Answer: C — both openings renew air so burning can continue.

B lasts a bit longer than A, but still goes out.

Air plays an important role in burning.

Fresh air must continuously be supplied for burning to continue.

1. Air has a role in burning.

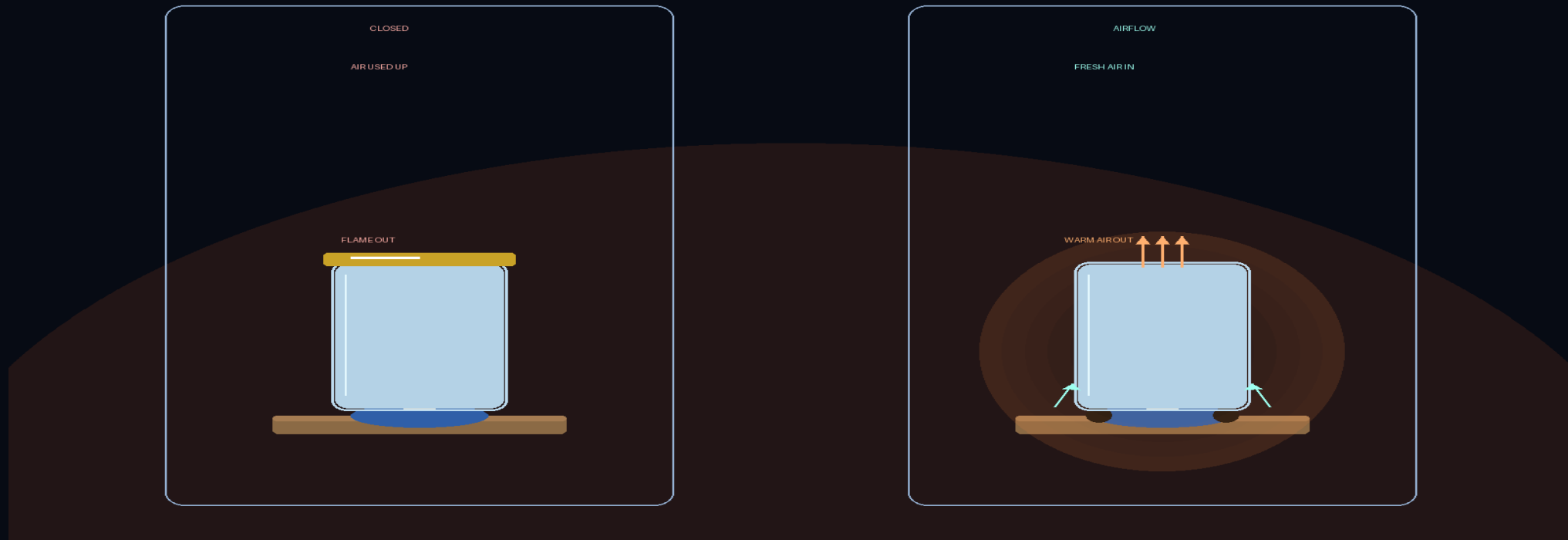
2. Fresh air must constantly flow in and out for burning to continue.

3. Where air is not renewed, things cannot continue burning.

REMEMBER

Investigation complete.

Air is invisible — but essential for a flame to keep burning.



Fuel + Oxygen + Heat → Heat + Light